

Grade 7
Unit 9
Lesson 7 Practice

Name _____
Date _____
Period _____

Determine if the expressions are equivalent for the given values.

	Value	$8x + 4x - 5$	$2(6x - 8) + 11$	Equivalent?
1.	$x = -4$			☐ yes ☐ no
2.	$x = 0$			☐ yes ☐ no
3.	$x = 0.5$			☐ yes ☐ no
4.	$x = \frac{1}{3}$			☐ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

- ☐ always
- ☐ sometimes
- ☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{5}(15 + 20y) + 2x$	$4\left(2x + y + \frac{3}{4}\right)$	Equivalent?
6.	$x = -2$ $y = 6$			☐ yes ☐ no
7.	$x = 0$ $y = 1$			☐ yes ☐ no
8.	$x = \underline{\hspace{1cm}}$ $y = \underline{\hspace{1cm}}$			☐ yes ☐ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(13 - 15) - \frac{3}{2}(4 + 6x)$	$-11x - 15$	☐ yes ☐ no
10.	$\frac{5}{7}(2x + 12x) - (6^2)$	$-36 + 10x$	☐ yes ☐ no